Genetics

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GENETICS

1. DNA STRUCTURE
Double helix (Watson & Crick)
Sugar-phosphate backbone

Purines: A, G
Pyrimidines: C, T
Human: 46 chromosomes

2. REPLICATION
Semi-conservative
Helicase, Primase, Polymerase, Ligase
5’->3’

3. TRANSCRIPTION
DNA -> mRNA
RNA polymerase
Promoter, Terminator
Introns removed, Exons joined
5’ cap, Poly-A tail

4. TRANSLATION
mRNA -> Protein
Ribosome, tRNA
Codon = 3 nucleotides
Start: AUG
Stop: UAA, UAG, UGA
64 codons -> 20 amino acids

5. MENDEL
Segregation
Independent assortment
Monohybrid: 3:1
Dihybrid: 9:3:3:1

6. VARIATIONS
Incomplete dominance
Codominance
Multiple alleles (ABO)
Polygenic
Sex-linked

7. MUTATIONS
Point: Silent, Missense, Nonsense
Frameshift
Chromosomal: Del, Dup, Inv, Trans

8. DNA TECHNOLOGY
PCR, Gel electrophoresis
CRISPR, Cloning
Genetic testing